



VISUAL ANALYTICS: EXPLAINABLE AI

AB MOSCA (THEY/THEM)



TODAY'S PLAN

- Why XAI Matters
- XAI and Visualization
- Open Areas

WHY XAI MATTERS

- **Power** describes the current configuration of structural privilege and structural oppression, in which some groups experience unearned advantages—because various systems have been designed by people like them and work for people like them—and other groups experience systematic disadvantages—because those same systems were not designed by them or with people like them in mind
- Who has structural privilege in data science / computer science?
- Who is structurally oppressed by data science / computer science?

WHY XAI MATTERS

- Name three examples of how AI is used for decision making
- What recourse do the people impacted by these decisions have if they disagree?

WHY XAI MATTERS

- One method by which we can reduce the power discrepancies in AI decision making is by revealing the algorithmic process behind the decisions, or in other words developing methods to explain AI output (Explainable AI)
- XAI focuses on topics including:
 - Trust
 - Interpretability
 - Explainability

XAI AND VISUALIZATION

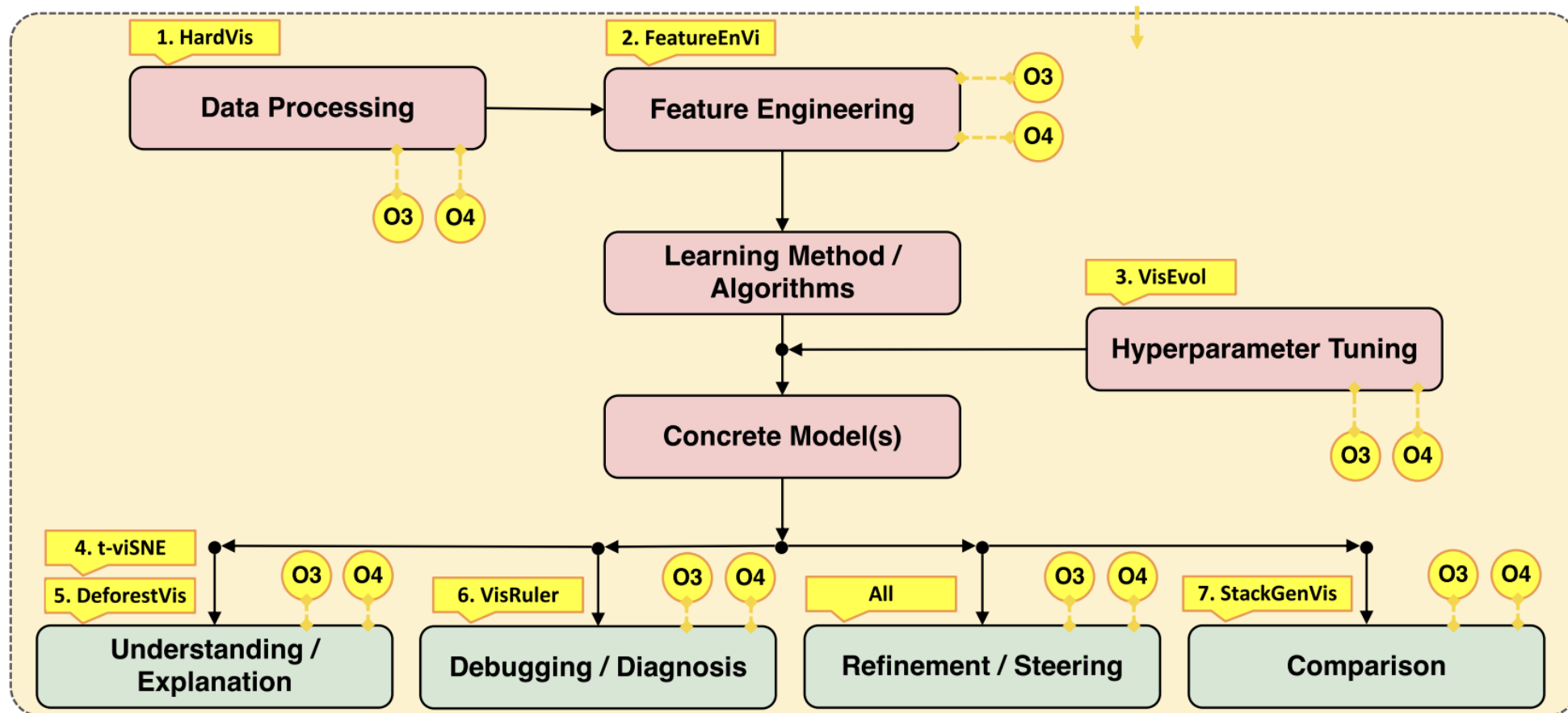
Table 4. Number of papers per year analyzed by each survey covered in this work.

	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Amershi et al. ²⁹	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	2	2	-	2	1	-	-	-	-	-
Choo and Liu. ³⁰	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	2	-	2	6	5
Dudley and Kristensson ³¹	-	-	-	-	-	-	-	-	-	1	-	1	-	1	-	-	1	3	2	3	4	4	7	14	8	1	-
Endert et al. ²⁷	-	-	-	-	-	-	-	-	-	-	1	-	2	-	1	-	9	3	3	7	5	5	5	2	4	-	-
Garcia et al. ³²	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	2	7	12	12	6
Hohman et al. ²⁰	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	3	4	10	12	8
Liu et al. ²⁸	-	-	-	-	-	-	-	-	1	-	-	1	-	1	-	1	-	1	4	3	2	2	1	-	-	-	-
Liu et al. ³³	1	-	1	-	1	-	1	-	-	1	-	-	1	1	-	-	1	1	1	1	3	2	4	5	7	4	-
Lu et al. ³⁴	-	-	-	-	-	2	-	1	1	2	1	1	1	-	-	-	1	1	2	3	-	2	11	1	1	-	-
Lu et al. ³⁵	-	-	-	-	-	-	-	1	1	-	1	-	-	-	-	2	1	1	6	7	7	2	3	3	4	3	-
Sacha et al. ³⁶	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	1	1	3	1	2	1	4	1	-	-	-
Seifert et al. ³⁷	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	3	1	12	9	7	1	-
Wang et al. ²⁶	-	-	-	-	-	1	-	-	-	-	-	-	2	-	-	-	-	-	-	-	-	1	2	-	-	-	2
Yu and Shi ³⁸	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	1	2	4	2	6	9
Zhang and Zhu ³⁹	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	5	-	5	12	2
Grün et al. ⁴⁰	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	1	2	1	1	7	4	1	-	-
Sacha et al. ⁴¹	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	1	1	2	1	-	-	-	-	-
Samek et al. ⁴²	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	1	-	-	-	4	4	5	5	-
Number of unique papers/year	1	0	1	0	1	3	1	2	3	4	3	4	6	5	1	4	15	18	27	28	32	24	74	58	68	62	32

ML: machine learning.

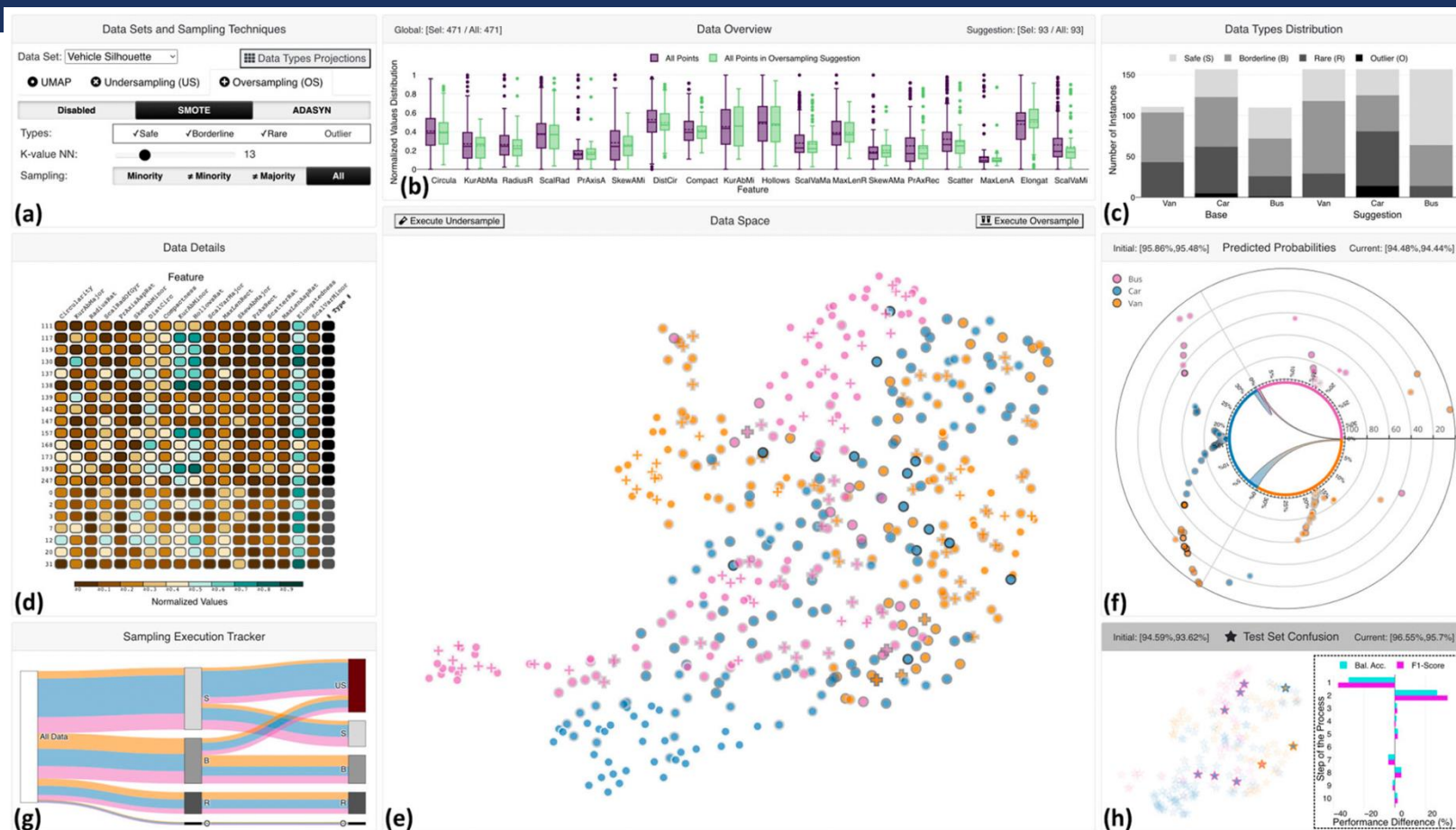
Before 2008, the frequency of papers in the subject is low (white and light blue colors). From 2008 until 2013, we can see an increase in the publications for interpretable ML with visualization (dark blue colors), reaching its peak in 2014. Since then, the annual frequency of related papers has stabilized.

XAI AND VISUALIZATION: DESIGN SPACE



XAI AND VISUALIZATION: DATA PROCESSING

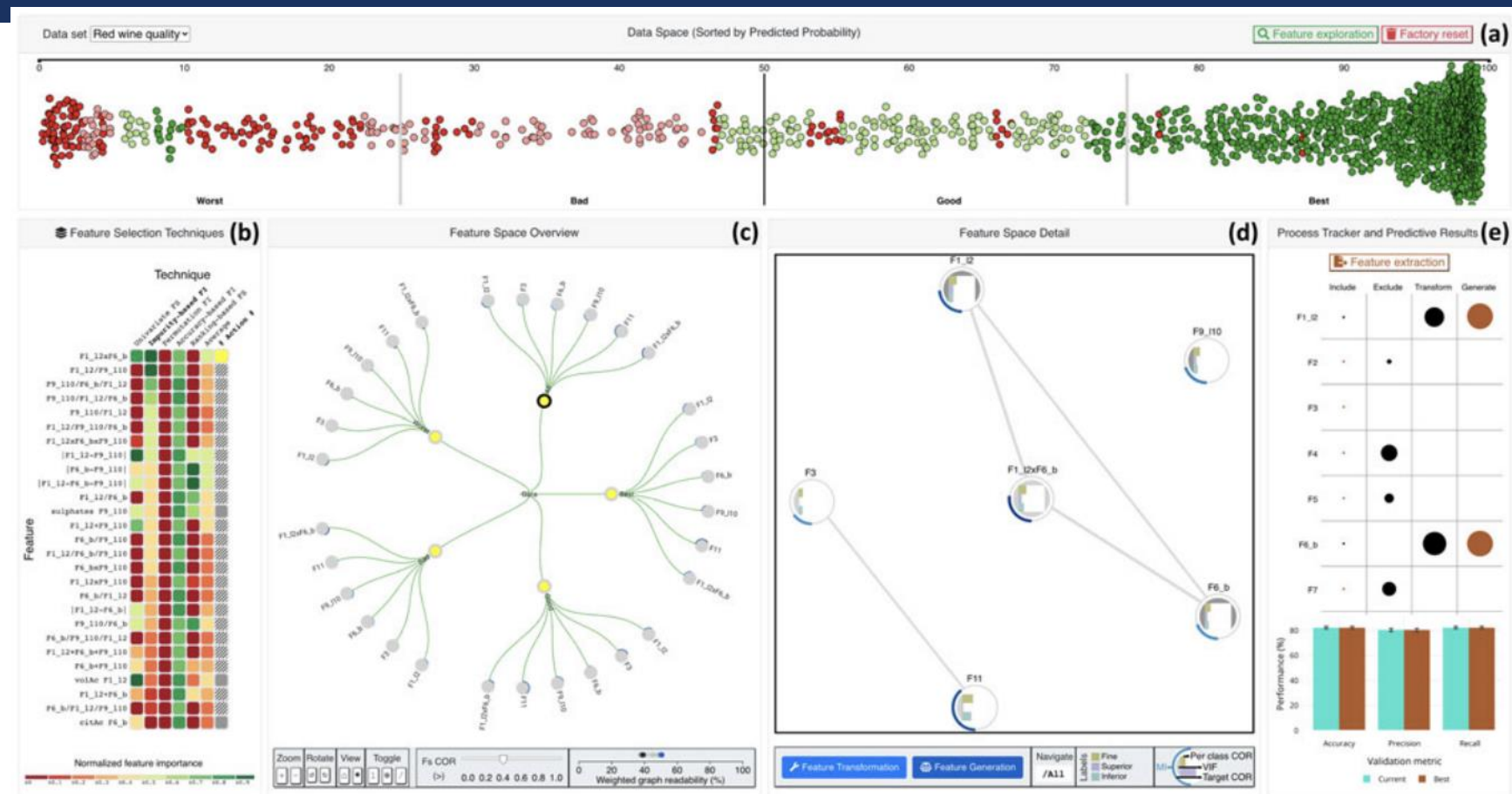
- HardVis
- <https://vimeo.com/772796696>



Chatzimpampas, et al. (2022). HardVis: Visual Analytics to Handle Instance Hardness Using Undersampling and Oversampling Techniques. 42. 10.1111/cgf.14726.

XAI AND VISUALIZATION: FEATURE ENGINEERING

- FeatureEnVi
- <https://vimeo.com/662536366>



A. Chatzimpampas, et al., "FeatureEnVi: Visual Analytics for Feature Engineering Using Stepwise Selection and Semi-Automatic Extraction Approaches," 2022, doi: 10.1109/TVCG.2022.3141040

XAI AND VISUALIZATION: HYPERPARAMETER TUNING

■ VisEvol

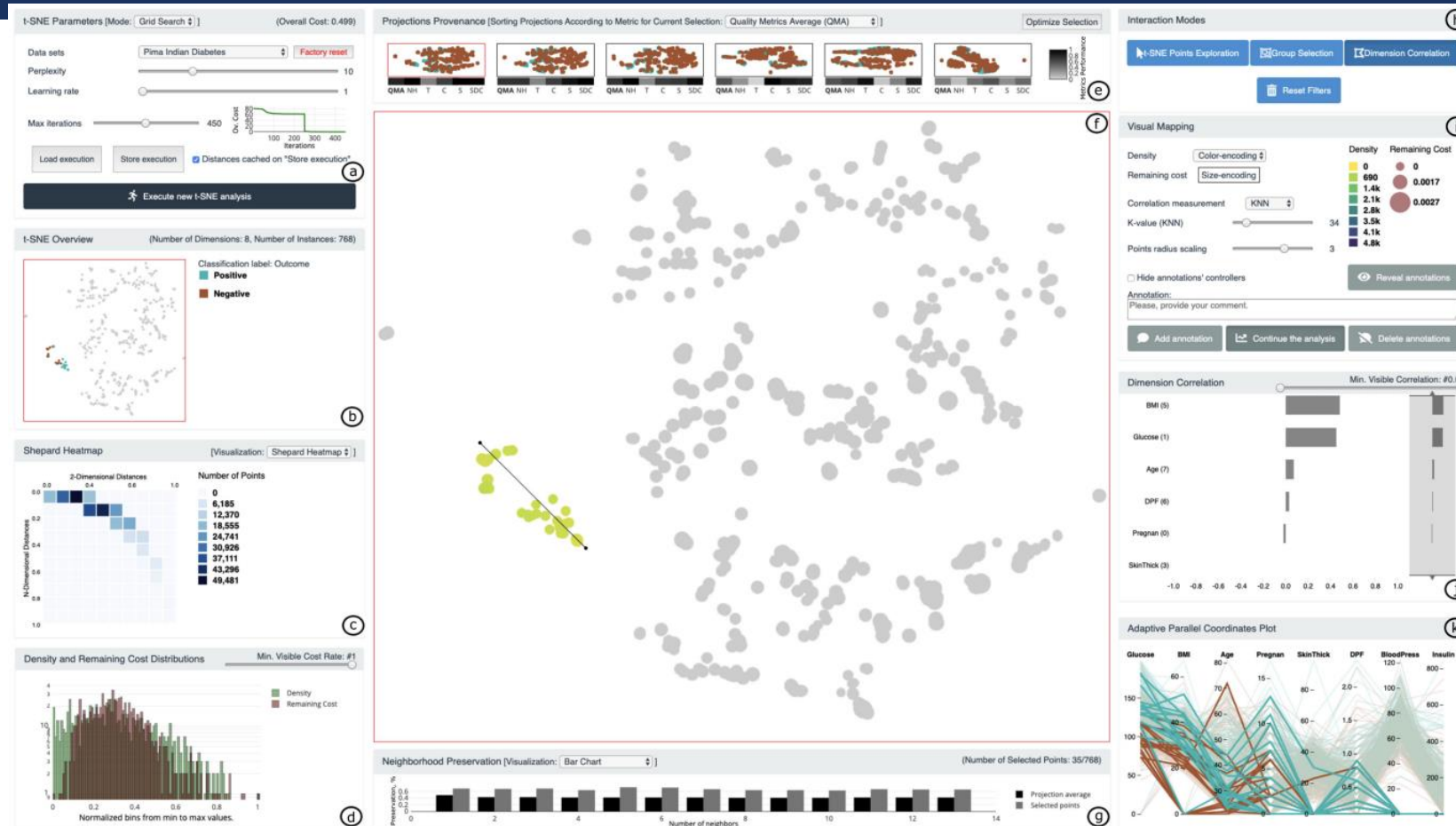
■ <https://diglib.eg.org/items/df6b539d-af1e-4a1a-b507-35a6d23c2d6f>



A. Chatzimpampas, "Visual Analytics for Explainable and Trustworthy Artificial Intelligence," 2025, doi: 10.1109/MCG.2025.3533806

XAI AND VISUALIZATION: MODEL UNDERSTANDING

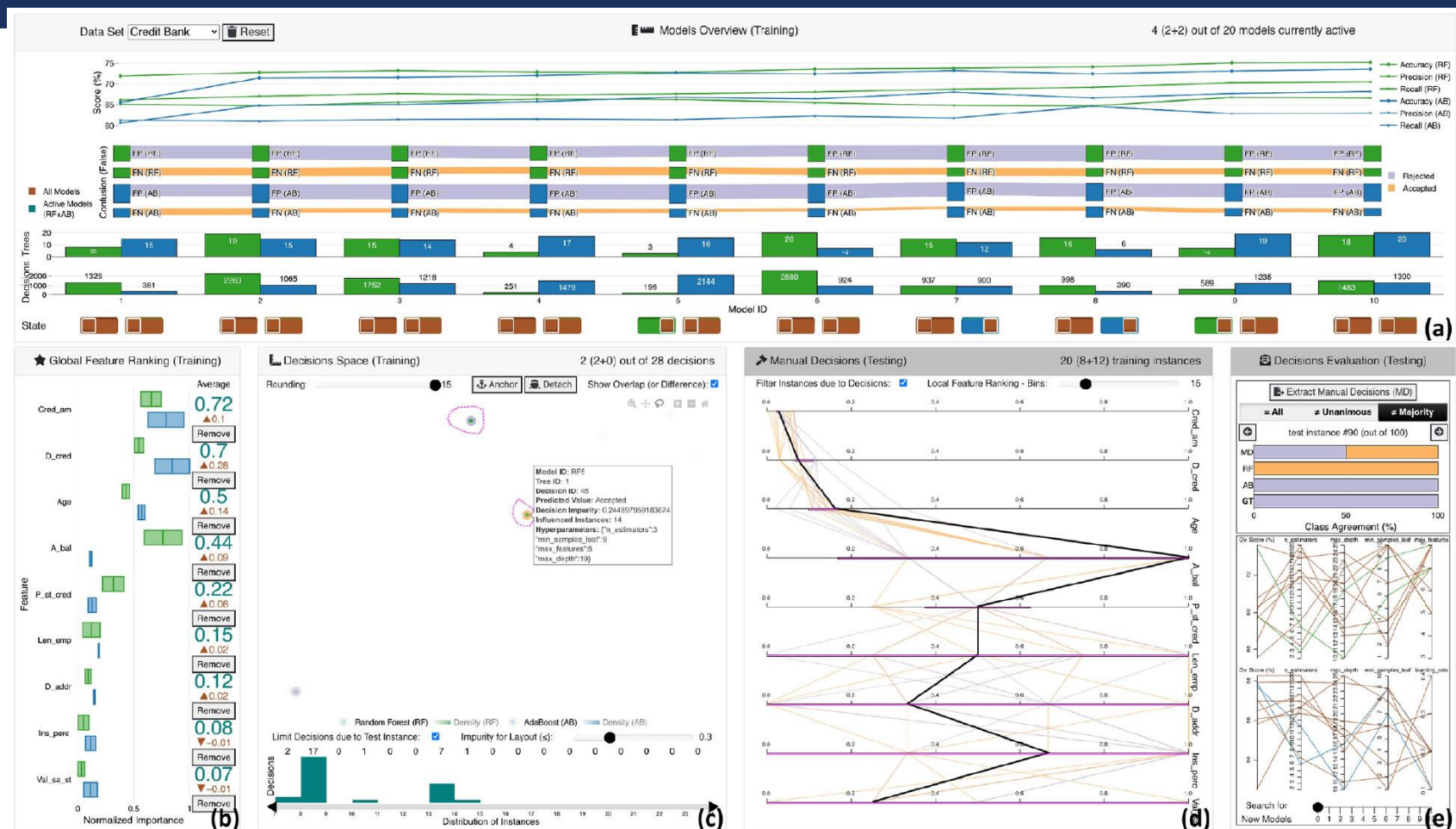
- t-viSNE
- <https://ieeexplore.ieee.org/document/9064929/media#media>



A. Chatzimpampas, et al. " t-viSNE: Interactive Assessment and Interpretation of t-SNE Projections," 2020, doi: 10.1109/TVCG.2020.2986996

XAI AND VISUALIZATION: MODEL DEBUGGING

- VisRuler
- <https://journals.sagepub.com/doi/10.1177/14738716221142005#supplementary-materials>

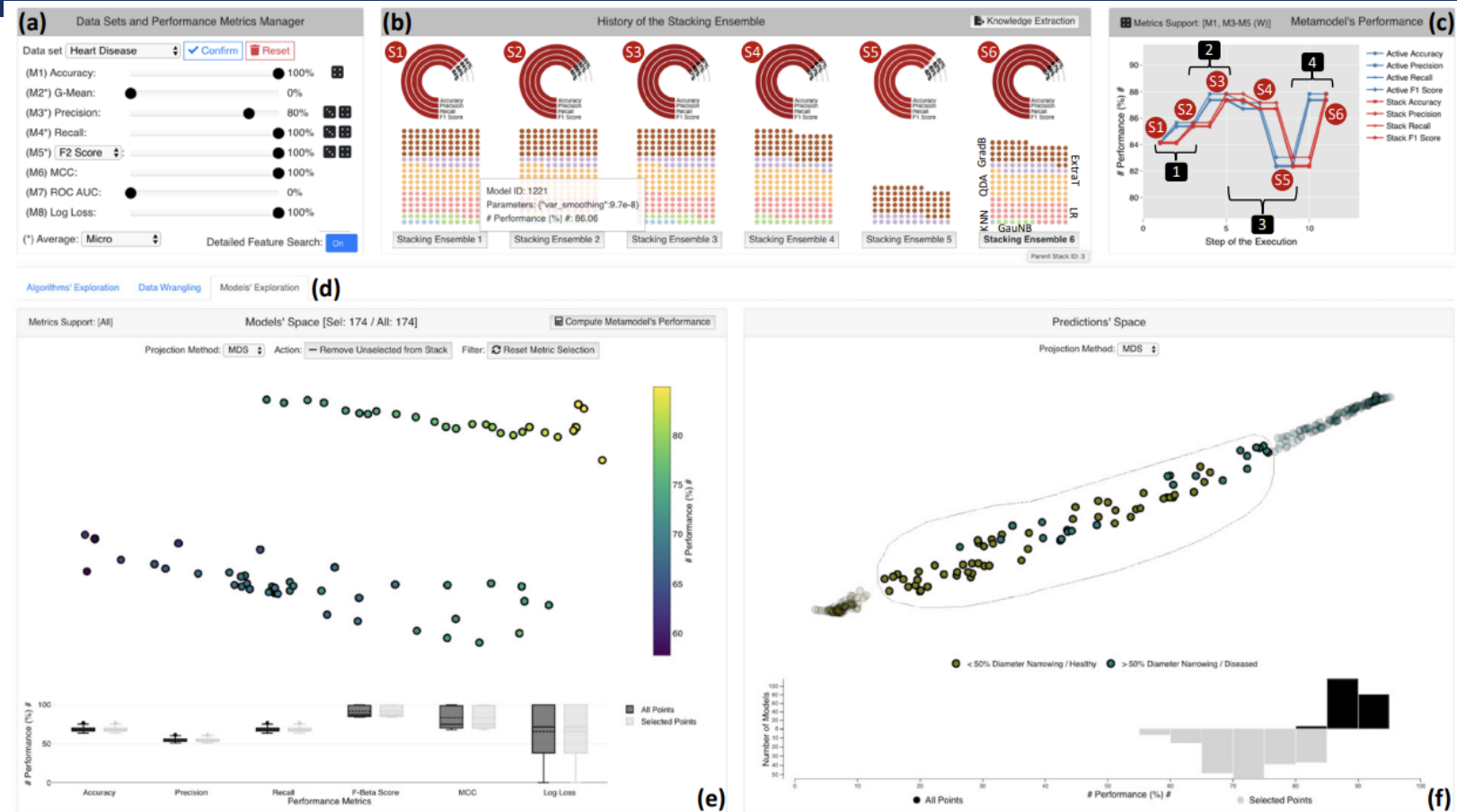


A. Chatzimpampas, et al."VisRuler:Visual analytics for extracting decision rules from bagged and boosted decision trees," 2023, doi: 10.1177/14738716221142005

XAI AND VISUALIZATION: MODEL COMPARISONS

■ StackGenVis

■ <https://ieeexplore.ieee.org/document/9222343/media#media>



A. Chatzimpampas, et al. " StackGenVis:Alignment of Data,Algorithms, and Models for Stacking Ensemble Learning Using Performance Metrics," 2021, doi: 10.1109/TVCG.2020.3030352

OPEN AREAS: OVER FOCUS ON NEURAL NETS

- The majority of XAI work in the VA community focuses on NNs for image data
- Out of 542 visualization papers surveyed on enhancing trust in AI, 345 primarily addressed NN models, while only 114 concentrated on ensembles.
- Although image analysis is undeniably an important research domain—especially in fields, such as medicine, where NNs excel with high-dimensional data—other application areas should not be overlooked
- Tabular data still represent the majority of datasets employed in AI workflows, where ensemble learning models often surpass NNs in performance

OPEN AREAS: UNDEREXPLORED OPPORTUNITIES

- VA dashboards for boosting and stacking ensemble learning methods are notably scarce
- Multilabel and regression problems are less explored compared to classification
- In the realm of unsupervised learning, association and pattern mining remain underrepresented
- Reinforcement learning also receives minimal attention
- Lastly, more emphasis is needed on addressing the needs of beginners and users with varying levels of experience who aim to analyze their data broadly

YOUR TURN!

- Navigate to: <https://trustmlvis.lnu.se/>
- Notice the “Target Group” filter option at the bottom of the page
- Inspect some tools for Beginners, Practitioners, and Developers
 - What similarities and differences do you see?
 - Do the tools you seem seem appropriate for the target group?
 - Sketch out how you would redesign a beginner tool for a practitioner and vice versa